

UME Seminar

Transitional regimes in incompressible wall-bounded shear flows

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Date and Time: 8th of June (2 – 3 pm)

Venue: Amphi 104 - Pôle Mécanique

Abstract

Wall-bounded flows are known to support turbulence even in regimes where the laminar flow is linearly stable. This can lead, at low enough Reynolds numbers, to a spatio-temporal competition between the two types of dynamics. Many questions about these mixed regimes stay open as of today. In particular, we would like to understand the mechanisms leading from homogeneous shear turbulence to regular laminar/turbulent patterns. In this talk we will review the latest developments on the topic, involving large-scale simulations, stability theory in a turbulent context, and phenomenological modelling attempts.

About the speaker

Yohann Duguet est chargé de recherche CNRS au LISN (Université Paris-Saclay). Spécialiste de la mécanique des fluides et de la transition vers la turbulence, ses travaux portent sur la dynamique des écoulements cisailés, les phénomènes d'intermittence laminaire-turbulente et l'application des outils des systèmes dynamiques à l'étude des écoulements complexes. Après une thèse à l'École Centrale de Lyon, puis des séjours postdoctoraux à l'University of Bristol et au KTH Royal Institute of Technology, il a rejoint le CNRS en 2009.



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